



HydroSovereign AI Engine

QGIS Plugin v6.01

Complete User Manual — 10 Chapters

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Security Scan	✓ Passed · 0 Critical · 0 Warnings · 76 files scanned
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Document Revision History

Version	Date	Author	Changes
1.0	March 2026	S.M.G. Alkedir	Initial draft
2.0	April 16, 2026	S.M.G. Alkedir	Added 13 tools, 5 algorithms, PR #289 approval
3.0	April 21,	S.M.G. Alkedir	Official publication (Plugin ID: 5040). Updated

	2026		installation to QGIS Plugin Repository. Security scan results. 21+ downloads on first day.
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Quick Start — 5-Minute Tutorial

Follow these 5 steps to go from installation to your first basin analysis:

Step 1 — Install from Official Repository (1 minute)

1. Open QGIS (version 3.16 or newer)
2. Plugins → Manage and Install Plugins → search 'HydroSovereign' → Install Plugin
3. Or go directly: https://plugins.qgis.org/plugins/hsae_qgis/ (Plugin ID: 5040)
4.  The HSAE v6.01 toolbar and menu appear immediately

Step 2 — Load the Basin Registry (10 seconds)

5. Click the  button in the HSAE toolbar
6. 26 basin points appear on the map — coloured by ATDI risk level

Step 3 — Apply ATDI Colour Map (5 seconds)

7. Click  TDI/ATDI Visualiser — basins recolour Green → Purple

Step 4 — Generate a WebGIS Map (30 seconds)

8. Click  WebGIS Map → save as .html → opens in browser

Step 5 — Export an ICJ Dossier (30 seconds)

9. Menu: HydroSovereign AI Engine v6.01 →  ICJ/PCA Dossier → save as .html

 Blue Nile (GERD) results: ATDI = 43.5% (Art. 7 UNWC), HIFD = 20.0%, NSE = 0.63, KGE = 0.74, CI = 0.44 (HIGH), P(Negotiation) = 58%

Chapter 1 — Overview & Introduction

1.1 What is the HSAE QGIS Plugin?

The HydroSovereign AI Engine QGIS Plugin v6.01 is officially published in the QGIS Plugin Repository (Plugin ID: 5040, approved April 21, 2026). It brings AI-powered transboundary water sovereignty analysis directly into QGIS desktop. It provides 13 interactive tools and 5 Processing Toolbox algorithms covering 26 globally contested transboundary river basins.

✓ Security scan passed: 0 Critical · 0 Warnings · 76 files scanned · Approved by QGIS reviewer
zimbogisgeek

1.2 Does It Require a Separate QGIS Profile?

No. Installs into any existing QGIS profile (version 3.16+). No separate profile required.

1.3 Plugin Components At a Glance

Component	Count	Access
Toolbar Tools	13 tools	HydroSovereign AI Engine v6.01 menu + toolbar
Processing Algorithms	5 algorithms	Processing Toolbox → HydroSovereign AI Engine v6.01
Dashboard Panel	1 dockable	Tool 11 toggles open/close
Basin Registry	26 basins	Built-in basins_50.json — 7 world regions
GEE Satellite Scripts	7 sensors	Tool 6 generates GEE JavaScript
GRDC Stations	10 stations	Tool 7 loads as point layer

Chapter 2 — Installation & Requirements

2.1 System Requirements

Component	Minimum	Recommended
QGIS	3.16 LTR	3.28 LTR or newer
Python	3.8 (bundled)	3.10+ (bundled with QGIS 3.28)
OS	Windows 10, macOS 10.15, Ubuntu 20.04	Windows 11, macOS 13, Ubuntu 22.04
Internet	Required for GEE scripts	Stable broadband
Extra Python Packages	None — all bundled with QGIS	—

2.2 Method A — QGIS Plugin Repository (Recommended)

Plugin ID: 5040 · https://plugins.qgis.org/plugins/hsae_qgis/

10. Open QGIS
11. Menu: Plugins → Manage and Install Plugins
12. In the search box, type: HydroSovereign
13. Select 'HydroSovereign AI Engine' (Plugin ID: 5040) from results
14. Click 'Install Plugin'
15.  The HSAE toolbar and menu appear immediately

 Security scan: 4/5 checks passed · 0 Critical · 0 Warnings · 76 files scanned (April 21, 2026)

2.3 Method B — Install from ZIP

16. Download: <https://github.com/saifeldinkhendir-coder/HydroSovereign-AI-Engine-HSAE-v601>
17. Plugins → Manage and Install Plugins → 'Install from ZIP' tab
18. Browse to HSAE_v601_QGIS_Plugin_FINAL_v2.zip → Install Plugin

 The ZIP internal folder must be `hsae_qgis/` — do not rename it. Method A (Plugin Repository) is always recommended.

2.4 Verifying the Installation

Location	What You Should See
QGIS Menu Bar	'HydroSovereign AI Engine v6.01' with 13 items
QGIS Toolbar	'HSAE v6.01' toolbar with 11 emoji buttons
Processing Toolbox	'HydroSovereign AI Engine v6.01' with 5 algorithms

2.5 Uninstalling

Plugins → Installed tab → Select 'HydroSovereign AI Engine' → Uninstall Plugin. All components removed cleanly.

Chapter 3 — Interface Overview

3.1 The HSAE Toolbar (11 buttons)

#	Emoji	Tool Name
1		Load Basin Registry
2		TDI/ATDI Visualiser
3		UNWC Legal Layer
4		Export Basin Data
5		Dashboard Dialog
6		GEE Script Generator
7		GRDC Stations Overlay
8		Conflict Index
9		Negotiation AI
10		WebGIS Map Generator
11		Basin Panel (dockable)

3.2 Menu-Only Tools

Tool	Description
12 —  ICJ/PCA Dossier	Full legal dossier for all 26 basins in ICJ/PCA format
13 —  About HSAE v6.01	Plugin metadata, links, DOI, and citation information

3.3 Processing Toolbox (5 Algorithms)

Processing → Toolbox → 'HydroSovereign AI Engine v6.01':

- **HSAE Indices:** ATDI Calculator, HIFD Calculator
- **HSAE Reports:** Basin Legal Report, Multi-Basin Comparison
- **HSAE Hydrology:** HBV-96 Calibration (SCE-UA)

Chapter 4 — Complete Tool Reference (13 Tools)

Tool 1 — Load Basin Registry

Purpose: Loads all 26 transboundary basins as a QGIS point layer with computed ATDI, HIFD, NSE, KGE, CI, P_negotiation attributes.

19. Click  in the HSAE toolbar
20. Layer 'HSAE Basin Registry (26 basins)' appears
21. Open Attribute Table (F6) to inspect all indices

 Apply Tool 2 immediately to colour basins by ATDI risk.

Tool 2 — TDI/ATDI Visualiser

Purpose: Applies graduated colour renderer based on ATDI values.

ATDI Range	Colour	UNWC Zone	Action
0–20%	 Green	Compliant	None
20–40%	 Yellow	Art. 7 NSH	Notify
40–55%	 Orange	Art. 9	Data sharing
55–70%	 Red	Art. 33	Dispute resolution
≥ 70%	 Purple	Art. 35	Emergency

Tool 3 — UNWC Legal Layer

Purpose: Creates a separate categorised layer showing UNWC 1997 legal compliance risk.

22. Click 
23. New layer 'HSAE UNWC Legal Risk' appears
24. Use Identify tool (I) to click any basin for legal details

Tool 4 — Export Basin Data

Purpose: Exports all 26 basins with computed indices to GeoJSON, Shapefile, or CSV.

Format	Best For	Geometry
GeoJSON	Leaflet, web maps	
Shapefile	ArcGIS, legacy GIS	
CSV	Excel, R, Python	Coordinates only

Tool 5 — Dashboard Dialog

Purpose: Opens per-basin analysis dialog. Select any of 26 basins → view all indices + triggered UNWC articles + negotiation strategy.

Tool 6 — 🗺️ GEE Script Generator (7 Sensors)

Purpose: Generates ready-to-run Google Earth Engine JavaScript for all 7 satellite sensors.

Sensor	Variable	Archive
GPM IMERG V07	Precipitation (mm/day)	2000–present
GRACE-FO MASCON RL06v4	TWS anomaly (cm)	2018–present
SMAP SPL3SMP	Soil moisture (m³/m³)	2015–present
Sentinel-1 GRD	SAR backscatter VV (dB)	2014–present
Sentinel-2 SR Harmonized	NDWI, NDVI	2015–present
ERA5 (ECMWF/GEE)	Air temperature (°C)	1979–present
Open-Meteo API	T, P, ET ₀ , SM	1940–present

Tool 7 — 📡 GRDC Stations Overlay

Purpose: Loads 10 key GRDC discharge monitoring stations as a styled QGIS point layer.

📄 GRDC observed discharge at grdc.bafg.de (free registration). HSAE calibration uses GloFAS ERA5 v4 proxy pending GRDC Request ID: 78949.

Tool 8 — ⚡ Conflict Index (26 Basins)

Purpose: Computes Composite Conflict Index for all 26 basins simultaneously, ranked from highest to lowest risk.

$$CI = 0.40 \times (ATDI/100) + 0.25 \times (dispute/4) + 0.20 \times (HIFD/100) + 0.15 \times (n_countries/6)$$

📊 Blue Nile (GERD): CI = 0.44 → HIGH

Tool 9 — 🤝 Negotiation AI

Purpose: Predicts negotiation success probability using GBM model trained on 478 historical TFDD/ICJ water dispute cases.

P(Success)	Strategy	UN Pathway
P ≥ 65%	Cooperative	Art. 8 Direct Negotiation
40–65%	Mediation	Art. 17 Third-Party Mediation
25–40%	PCA Arbitration	Art. 33 Dispute Resolution
P < 25%	ICJ Referral	International Court of Justice

Tool 10 — WebGIS Map Generator

Purpose: Generates standalone interactive HTML map (Leaflet.js). Works offline. All 26 basins plotted, styled by ATDI risk, fully interactive.

 The HTML file works in any browser without installation. Can be shared directly.

Tool 11 — Basin Panel (Dockable)

Purpose: Toggles dockable real-time dashboard panel. Click once to open, again to hide, again to reopen.

Tool 12 — ICJ/PCA Dossier (Menu Only)

Purpose: Exports complete structured legal dossier for all 26 basins in ICJ/PCA/ITLOS submission format.

 For formal international proceedings, the dossier must be reviewed by qualified water law experts. HSAE provides indicator-based assessments, not legal advice.

Tool 13 — About HSAE v6.01 (Menu Only)

Displays: Plugin ID: 5040 · Version 6.0.1 · DOI: 10.5281/zenodo.19180160 · Security scan: 0 critical issues · All relevant links.

Chapter 5 — Processing Toolbox Algorithms (5 Algorithms)

Algorithm 1 — ATDI Calculator

ID: atdicalculator **Group:** HSAE Indices

Parameter	Default	Description
Runoff Coefficient	0.38	Basin runoff ratio (GERD=0.38)
Dam Capacity (BCM)	74.0	Reservoir storage in Billion m ³
Number of Countries	3	Riparian states
Dispute Level (1–5)	4	TFDD/ICOW conflict intensity

 GERD: RC=0.38, Cap=74 BCM, Countries=3, Dispute=4 → ATDI = 43.5%

Algorithm 2 — HIFD Calculator

ID: hifdcalculator **Group:** HSAE Indices

Same parameters as ATDI. Output: HIFD (%) — volumetric annual flow deficit.

 GERD: HIFD = 20.0% (20% of natural downstream flow estimated as withheld)

Algorithm 3 — Basin Legal Report

ID: basinreport **Group:** HSAE Reports

Generates complete UNWC 1997 legal assessment as TXT or HTML.

 GERD report: ATDI=43.5% → Arts. 5, 7, 9, 33 triggered. Recommendation: Art. 17 mediation.

Algorithm 4 — HBV-96 Calibration (SCE-UA)

ID: hbv96calibration **Group:** HSAE Hydrology

Runs HBV-96 with Shuffled Complex Evolution (SCE-UA) optimisation. Returns NSE, KGE, calibrated parameters.

 Blue Nile default: NSE = 0.63, KGE = 0.74 (proxy-validated against GloFAS ERA5 v4)

 Ground-truth calibration pending GRDC Request ID: 78949 (El Diem station, 174,000 km²).

Algorithm 5 — Multi-Basin Comparison

ID: multibasincomparison **Group:** HSAE Reports

Generates ranked comparative report for multiple basins. Output: CSV + HTML.

Chapter 6 — The 26 Registered Basins

Colour: ● ATDI<20% · ● 20–40% · ● 40–55% · ● 55–70% · ● ≥70%

#	Basin	Dam	Cap BCM	Countries	ATDI%	HIFD%	Risk
1	Blue Nile	GERD	74	3	53.5%	35.7%	●
2	Nile	Aswan	162	2	33.1%	32.2%	●
3	Nile	Roseires	7.4	2	28.4%	24.1%	●
4	Zambezi	Kariba	180.6	2	31.1%	29.2%	●
5	Congo	Inga	2	8	49.1%	28.5%	●
6	Niger	Kainji	15	9	52.4%	37.2%	●
7	Euphrates	Atatürk	48.7	3	44.5%	34.9%	●
8	Tigris	Mosul	11.1	2	32.5%	29.1%	●
9	Vakhsh	Nurek	10.5	5	56.8%	35.5%	●
10	Syr Darya	Toktogul	19.5	4	48.2%	31.4%	●
11	Indus	Tarbela	13.7	2	30.5%	26.7%	●
12	Brahmaputra	Subansiri	2.4	3	29.1%	18.6%	●
13	Ganges	Farakka	0.3	2	27.3%	22.5%	●
14	Mekong	Xayaburi	7.4	6	56.5%	32.7%	●
15	Salween	Myitsone	62	3	41.2%	30.8%	●
16	Yangtze	Three Gorges	39.3	1	18.4%	22.8%	●
17	Danube	Iron Gates	2.4	10	40%	25.4%	●
18	Rhine	Various	0.5	9	27.1%	18.6%	●
19	Dnieper	Kakhovka	18.2	3	44.8%	31.2%	●
20	Amazon	Belo Monte	250	8	47%	31.9%	●
21	Paraná	Itaipu	29	2	17.5%	21.4%	●
22	Orinoco	Guri	135	2	22.6%	24.3%	●
23	Colorado	Hoover	36.7	2	33.4%	32.5%	●
24	Columbia	Grand Coulee	9.7	2	19.8%	23.1%	●

4							
2 5	Rio Grande	Amistad	5.8	2	26.4%	27.8%	
2 6	Murray-Darling	Hume	3	2	21.3%	19.7%	

Chapter 7 — Scientific Indices Reference

7.1 ATDI — Alkedir Transparency Deficit Index

Step 1: $I_{adj,i} = I_{nat,i} - ET_{up,i}$ Step 2: $TDI_i = clip[(I_{adj,i} - Q_{obs,i}) / (I_{adj,i} + \epsilon), 0, 1]$ Step 3: $ATDI = (1/N) \times \sum TDI_i \times 100\%$

 RMSE = 4.1% (calibrated against 14 published basin values)

7.2 HIFD — Human-Induced Flow Deficit

$HIFD = [(Q_{nat} - Q_{obs}) / Q_{nat}] \times 100\%$

 GERD 2024: HIFD = 20.0% — 20% of natural flow estimated as withheld

7.3 Conflict Index

$CI = 0.40 \times (ATDI/100) + 0.25 \times (dispute/4) + 0.20 \times (HIFD/100) + 0.15 \times (n_countries/6)$

7.4 HBV-96 Performance (NSE & KGE)

Metric	NSE	KGE
Very Good	≥ 0.75	≥ 0.75
Good	0.65–0.75	0.65–0.75
Satisfactory	0.50–0.65	0.50–0.65
HSAE Blue Nile (proxy)	0.63 	0.74 

 Thresholds from Moriasi et al. (2007). NSE=0.63 proxy-validated vs GloFAS ERA5 v4 reanalysis.

Chapter 8 — Google Earth Engine Data Sources

Source	Variable	Resolution	Archive
GPM IMERG V07	Precipitation (mm/day)	0.1°	2000–
GRACE-FO MASCON	TWS anomaly (cm)	0.5°	2018–
SMAP SPL3SMP	Soil moisture (m ³ /m ³)	10 km	2015–
Sentinel-1 GRD	SAR backscatter VV (dB)	10 m	2014–
Sentinel-2 SR	NDWI, NDVI	10 m	2015–
ERA5 (ECMWF)	Air temperature (°C)	0.25°	1979–
Open-Meteo API	T, P, ET ₀ , SM	0.25°	1940–

Chapter 9 — Troubleshooting & FAQ

Problem	Solution
Plugin not found in QGIS Plugin Manager	Update QGIS plugin list (Settings → Plugin Manager → scroll to top → 'Reload all repositories'). Then search 'HydroSovereign'.
Plugin missing after installation	Restart QGIS. Then: Plugins → Manage → check 'HydroSovereign AI Engine' checkbox is ticked.
Error loading plugin on QGIS start	Update QGIS to ≥ 3.16. The plugin requires Python 3.8+ bundled with QGIS.
Processing algorithms missing	Enable Processing plugin: Plugins → Manage → tick 'Processing'. Open Toolbox (Ctrl+Alt+T).
TDI Visualiser: 'Select a basin layer first'	Run Tool 1 first, single-click the basin layer in Layers panel, then run Tool 2.
WebGIS map shows blank	Reinstall plugin. Ensure JavaScript enabled in browser.
ICJ Dossier file empty	Choose writable folder (Desktop, Documents). Avoid system-protected folders.
NSE/KGE seem low	NSE=0.63, KGE=0.74 are proxy-validated results. Post-calibration with GRDC Q_obs (Request ID: 78949) expected NSE > 0.75.
Plugin ID 5040 not shown in About dialog	Update to v6.0.1 via Plugin Manager → Upgradeable tab.

Chapter 10 — Citation & References

10.1 Primary Citation

When using the HSAE QGIS Plugin in research or publications:

```
@software{alkedir2026hsae_qgis, author = {Alkedir, Seifeldin M.G.}, title = {{HydroSovereign AI Engine QGIS Plugin v6.01}}, year = {2026}, publisher = {QGIS Plugin Repository}, note = {Plugin ID: 5040. Approved April 21, 2026.}, url = {https://plugins.qgis.org/plugins/hsae_qgis/}, doi = {10.5281/zenodo.19180160}, orcid = {0000-0003-0821-2991} }
```

10.2 Primary Paper

Alkedir, S.M.G. (2026). HydroSovereign AI Engine (HSAE) v6.01. SoftwareX (under review). DOI: 10.5281/zenodo.19180160

10.3 Key References

- Nash & Sutcliffe (1970). Journal of Hydrology, 10(3), 282–290.
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- Bergström (1992). HBV model. SMHI Reports Hydrology No. 4.
- Duan et al. (1993). SCE-UA. Water Resources Research, 28(4), 1015–1031.
- Huffman et al. (2023). GPM IMERG V07. doi:10.5067/GPM/IMERG/3B-HH/07
- United Nations (1997). UNWC. UN Doc. A/51/869.

Appendix A — Glossary

Term	Definition
ATDI	Alkedir Transparency Deficit Index — daily fraction of natural downstream entitlement withheld (0–100%)
HIFD	Human-Induced Flow Deficit — annual volumetric flow reduction from dam operations (0–100%)
CI	Conflict Index — composite risk score 0–1
UNWC	UN Watercourses Convention 1997
GEE	Google Earth Engine — cloud geospatial analysis platform
HBV-96	Rainfall-runoff model (Bergström 1992)
SCE-UA	Shuffled Complex Evolution — global optimisation algorithm (Duan 1993)
NSE	Nash-Sutcliffe Efficiency (≥ 0.75 = very good)
KGE	Kling-Gupta Efficiency (≥ 0.75 = very good)
GRDC	Global Runoff Data Centre — BfG Koblenz
EnKF	Ensemble Kalman Filter (n=200 members in HSAE Digital Twin)
GBM	Gradient Boosting Machine — Negotiation AI model (478 cases)
TFDD	Transboundary Freshwater Dispute Database — Oregon State University
ICJ	International Court of Justice
PCA	Permanent Court of Arbitration

Appendix B — All Links & Resources

Resource	URL
QGIS Plugin Repository (ID: 5040)	https://plugins.qgis.org/plugins/hsae_qgis/
GitHub Repository	https://github.com/saifeldinkhedar-coder/HydroSovereign-AI-Engine-HSAE-v601
Live Streamlit App	https://hydrosovereign-ai-engine-hsae-v601-6euz2zxcmerkzvgordmvxf.streamlit.app
Python Package (PyPI)	https://pypi.org/project/hydrosovereign/ (pip install hydrosovereign)
Zenodo Archive (DOI)	https://doi.org/10.5281/zenodo.19180160
ORCID	https://orcid.org/0000-0003-0821-2991
Bug Reports	https://github.com/saifeldinkhedar-coder/HydroSovereign-AI-Engine-HSAE-v601/issues
GRDC Data Portal	https://grdc.bafg.de (free registration)
Google Earth Engine	https://code.earthengine.google.com
QGIS Download	https://qgis.org/en/site/forusers/download.html
SoftwareX Paper	Under Review 2026 · DOI: 10.5281/zenodo.19180160